

## SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

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### COURSE OUTCOME COVERED

**CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

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Time Duration: 1 Hour

Total Marks:40

QUESTIONS: 4

Annexure A

### CASE STUDY

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#### Code of Conduct:

I M. Arshini 190424322 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Arshini

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## Annexure A

### CASE STUDY

**Student 15: M ARSHINI | Register Number: 192424322**

#### Question 1

##### **Case Study - Vehicle Fault Diagnosis System**

An automobile service center is designing a rule-based Logical Agent for preliminary vehicle fault diagnosis. The system's knowledge base contains:

- i. If there is EngineWarningLight AND UnusualNoise  $\rightarrow$  Possible EngineFailureRisk
- ii. If there is EngineWarningLight AND LowOilPressure  $\rightarrow$  Possible OilLeak
- iii. If there is UnusualNoise AND OverheatingSign  $\rightarrow$  Possible CoolantSystemFailure
- iv. Facts about Vehicle X: EngineWarningLight = True, UnusualNoise = True, OverheatingSign = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible faults for Vehicle X. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Vehicle X has CoolantSystemFailure.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

#### Question 2

##### **Case Study - Hospital Emergency Ward Routing Agent**

A hospital uses a FOL-based knowledge base to manage robot routing priority in emergency corridors:

- I. for-all x: Patient(x) AND CriticalCondition(x)  $\rightarrow$  PriorityCorridor(x) [Rule 1]
- II. for-all x: PriorityCorridor(x)  $\rightarrow$  CorridorCleared(x) AND NOT NormalQueue(x) [Rule 2]
- III. for-all x for-all y: Patient(x) AND WaitingBehind(x,y) AND CorridorCleared(y)  $\rightarrow$  Hold(x) [Rule 3]
- IV. Facts: Patient(PatientA), CriticalCondition(PatientA), Patient(PatientB), WaitingBehind(PatientB, PatientA)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.
- (b) Using the Resolution method, prove that 'Hold(PatientB)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous critical patients. Identify what additional rules or facts are needed and justify logically.

#### Question 3

##### **Case Study - Weather-Based Flight Scheduling Expert System**

An airline scheduling logical agent operates with the following propositional knowledge base (KB):

- 1) P1: StormWarning  $\rightarrow$  FlightDelayNeeded
- 2) P2: FlightDelayNeeded AND InternationalRoute  $\rightarrow$  NotifyPassengers



- 3) P3: NOT NotifyPassengers AND InternationalRoute  $\rightarrow$  ComplianceRisk  
4) Facts: StormWarning = True, InternationalRoute = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication  $\rightarrow$  move NOT inward  $\rightarrow$  distribute).
- (b) Using Resolution Refutation, prove the theorem 'NotifyPassengers'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'StormWarning' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'ComplianceRisk' still follow? Apply appropriate reasoning patterns to justify.

#### Question 4

#### **Case Study - Cave Exploration Knowledge Agent**

An AI agent exploring an underground cave grid perceives: Draft at [1,3], Glow at [1,1], Sparkle at [2,3]. The agent uses a propositional KB with these rules:

- 1) Draft[1,3]  $\rightarrow$  Pit is adjacent to cell [1,3]
- 2) Glow[1,1]  $\rightarrow$  LavaVent is adjacent to cell [1,1]
- 3) Sparkle[x,y]  $\rightarrow$  Treasure is at [x,y]
- 4) NOT Pit[x,y] AND NOT LavaVent[x,y]  $\rightarrow$  Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Pit and LavaVent. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1]  $\rightarrow$  [1,3]  $\rightarrow$  [2,3] is safe for the agent to collect the Treasure. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

# vehicle fault Diagnosis system

## a) Propositional logic representation

$E$  :- engine warning light on

$N$  :- unusual noise is present

$O$  :- overheating sign is present

$L$  :- Low oil pressure is present

$ER$  :- Possible engine failure risk

$OL$  :- possible oil leak

$CF$  :- possible coolant system failure

Rules :-

$R_1: (E \wedge N) \rightarrow ER$

$R_2: (E \wedge L) \rightarrow OL$

$R_3: (N \wedge O) \rightarrow CF$

Facts about vehicle  $x$  :-

$E = \text{True}$ ,  $N = \text{True}$ ,  $O = \text{True}$

## b) forward chaining :-

Initial facts :-  $E, N, O$

Step 1 :- check  $R_1: (E \wedge N) \rightarrow ER$

$E = \text{True}$  and  $N = \text{True}$

$\therefore$  Rule  $R_1$  fires

New fact deduced :-  $ER$

Step 1:-

check R2: (E, N, L)  $\rightarrow$  OL

E = True but L = unknown

$\therefore$  Rule R2 cannot fire

Step 2:-

check R3: (N, N, O)  $\rightarrow$  CF

N = True and O = True

$\therefore$  Rule R3 fires

New fact derived: CF

Step 4:- No new rules can fire

Final derived facts: E, N, O, ER, CF

possible faults for vehicle x.

c) Backward chaining to prove CF

Goal: CF

Rule supporting CF: R3: (N, N, O)  $\rightarrow$  CF

Subgoals: N, O



$\therefore$  Goal vehicle x has coolant system failure is



# Hospital Emergency ward Routing Agent

a) unification for rule 1:

Rule 1:  $\forall x: \text{Patient}(x) \wedge \text{critical condition} \rightarrow \text{priority corridor}(x)$

Given facts

F1:  $\text{patient}(\text{patient A})$

F2:  $\text{critical condition}(\text{patient A})$

Step 1:- match  $\text{patient}(x)$  with  $\text{patient}(\text{patient A})$

$$\theta_1 = \{x / \text{patient A}\}$$

Step 2:- Apply  $\theta_1$  to critical condition (patient A)

critical condition (patient A) matches F2

$$\text{MIGU } \theta = \{x / \text{patient A}\}$$

$\therefore$  from rule 1 and facts we derive priority corridor (patient A)

b) Resolution to prove hold (patient B)

1) converts to CNF

from rule 1:-  $\neg \text{patient}(x) \vee \neg \text{critical condition}(x) \vee \text{priority corridor}(x)$

from rule 2:-  $\neg \text{priority corridor}(x) \vee \text{corridor cleaned}(x)$   
 $\neg \text{priority corridor}(x) \vee \neg \text{normal queue}$

Facts:-

F<sub>1</sub>: patient (patient A)

F<sub>2</sub>: critical condition (patient A)

F<sub>3</sub>: patient (patient B)

F<sub>4</sub>: waiting Behind (patient B, patient A)

Wanted goal:-  $\neg \text{Hold}(\text{patient B})$

2) Resolution steps:-

1) from Rule 1 with F<sub>1</sub> and F<sub>2</sub>

Priority Corridor (patient A)

2) from rule 2 with (1)

Corridor cleared (patient A)

3) from Rule 2 with (1)

$\neg \text{Normal Queue}(\text{patient A})$

4) from Rule 3 with F<sub>3</sub>, F<sub>4</sub> and (2)

Hold patient (B)

5) from (4) and  $\neg \text{Hold}(\text{patient B})$

□ Empty (clause)

$\therefore \text{Hold}(\text{patient B})$  is true.



Resolution Refutation - prove Notify passengers

Goal:-

Notify passengers

Negate the Goal:-

7 Notify passengers

From the facts

storm warning

using P1:-

7 storm warning  $\vee$  flight

Delay needed

Therefore

flight delay needed

using P2:-

7 flight delay needed  $\vee$  7 International route

With

flight delay needed, International route

We derive

Notify passengers

But also we needed:

7 Notify passengers

c) If storm warning is removed

Initially

International route = true

But there is no storm warning.

Therefore:-

Notify passengers and

7 Notify passengers

Therefore P1 cannot fire

storm warning  $\rightarrow$

flight delay needed

so, flight delay needed is unavailable P2, cannot derive  
Also P3 requires.

7 Notify passengers

But this not given as a fact

Therefore

compliance Risk cannot be derived.



## 2) Weather Based flight scheduling expert system

Given KB:-

P1 :- storm warning  $\rightarrow$  flight delay needed

P2 :- Flight delay needed AND International route  $\rightarrow$  notify passengers

P3 :- NOT notify passengers AND International Route  $\rightarrow$  compliance risk

P4 :- facts .. storm warning = true,  
International route = true,

Step 1:- P1:-

storm warning  $\rightarrow$  Flight delay needed

Remove implication

~~storm warning~~  $\rightarrow$  flight delay needed

Step 2:- P2

Flight delay needed  $\wedge$  International route

CNIF:-

$\neg$  flight delay needed  $\vee$   $\neg$  International route

Step 2:- P3

$\neg$  notify passengers  $\wedge$  International route

CNIF:-

notify passengers  $\vee$  compliance risk

# cave exploration knowledge Agent

Given percepts

Draft at [1,3]

Glow at [1,1]

Sparkle at [2,3]

Rules:-

Draft [1,3]  $\rightarrow$  pit is adjacent to [1,3]

Glow [1,1]  $\rightarrow$  Lava vent is adjacent to [1,1]

Sparkle [x,y]  $\rightarrow$  treasure [x,y]

NOT pit [x,y] AND NOT Lava vent [x,y]  $\rightarrow$  safe [x,y]

a) Belief state and modus ponens

Draft at [1,3]

Glow [1,1]

Sparkle [2,3]

Rule 1:-

Draft [1,3]  $\rightarrow$  pit adjacent to [1,3]

Therefore,

pit is adjacent to [1,3]

Rule 2:-

Glow [1,1]  $\rightarrow$  Lava vent adjacent to [1,1]

Rule 3:-

Sparkle [x,y]  $\rightarrow$  treasure [x,y]

for  $x=2, y=3$ ;

Sparkle [2,3]  $\rightarrow$  treasure [2,3]

therefore

Treasure is at [2,3]



possible location of pit and Lavavent

pit:-

Draft is at  $[1,3]$

Adjacent cells are:

$[1,2]$   $[2,3]$ ,  $[1,4]$

So,

possible pit locations =  $[1,2]$   $[2,3]$   $[1,4]$

Lavavent:-

Grow is at  $[1,1]$

So,

possible Lavavent locations =  $[1,2]$ ,  $[2,1]$

check the path:-

path

$[1,1] \rightarrow [1,3] \rightarrow [2,3]$

we know

sparkle  $[2,3] \rightarrow$  treasure  $[2,3]$

so, treasure is present at  $[2,3]$

for a cell to be safe

NOT pit AND NOT Lavavent  $\rightarrow$  safe

But the knowledge bases does not give enough informations  
to prove that all three path cells have no pit or  
Lavavent.

## RUBRICS FOR CALCULATION

| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                     | Level 1 – Beginning                                  | Marks (100) |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|------------------------------------------|------------------------------------------------------|-------------|
| Understanding of Case   | 25        | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding: problem not identified correctly | 24          |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts          | Fails to apply relevant concepts                     | 24          |
| Analysis                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth        | Weak or no analysis; lacks reasoning                 | 19          |
| Solution Design         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification  | Incorrect or no solution provided                    | 19          |
| Clarity                 | 10        | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure            | Poor clarity; difficult to understand                | 9           |

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